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Metadata

Title*

Feeling the Same, Seeing the Same: Actual and Perceived Affect Similarity in Romantic Relationships

Description*

Decades of work in social and personality psychology posit that similarity between romantic partners benefits their relationship (Anderson et al., 2003; Luo, 2017; Murray et al., 2002). Similarity is argued to foster increased mutual understanding and closeness (Berlamont et al., 2024; Luo, 2017), and to signal that partners are “kindred spirits” (Murray et al., 2002). Empirical findings, however, vary across domains of similarity (e.g., traits, values, demographics, attitudes, and preferences), and results are generally small and mixed (From et al., 2025; Luo, 2017; Tidwell et al., 2013).

For similarity in affect, i.e. feeling the same or having similar emotional reactions to events, research findings are equally inconclusive regarding its association with relationship satisfaction, closeness, and responsiveness (Anderson et al., 2003; Gonzaga et al., 2007, 2010; Sels et al., 2020). Furthermore, the implications of affect similarity may depend on the valence of the affect. Similarity in positive affect (PA) is in general believed to predict relationship satisfaction for dating and married couples (Gonzaga et al., 2007, 2010), although other studies have failed to find consistent benefits (From et al., 2025). Similarity in negative affect (NA) on the other hand, especially in contexts of conflict or stress, appears to be associated with lower relationship quality (Brown et al., 2022; Gottman & Levenson, 2000; Levenson & Gottman, 1983).

Conceptually, PA and NA similarity might matter for different reasons. Specifically, PA similarity could signal common appraisals (‘we interpret the situation similarly’), interpersonal attunement (‘you get me’), or smooth co-regulation (‘we can manage this together’), all of which are plausible pathways to intimacy (Reis & Gable, 2015). Conversely, NA similarity may for example reflect increased internalized distress through co-rumination (‘we keep rehashing it together’; Rose, 2002), negative reciprocity and co-dysregulation during conflict (‘we escalate each other’; Butler, 2015; Gottman & Levenson, 2000), or the transmission of stress between partners ('your stress becomes mine'; Westman, 2001). Interpersonal theories of emotion dynamics emphasize that partners’ emotions are not independent streams: they can converge or escalate in response to shared events, react to each other, and become coupled through interpersonal adaptive or maladaptive emotion regulation processes (Butler, 2015; Butler & Randall, 2013). This view implies that similarity is not static but a fluctuating, time-varying process, that can best be captured when both partners are measured repeatedly in daily life or during interaction contexts (Bolger & Laurenceau, 2013; Kenny et al., 2020).

A possible explanation for the mixed findings around the association between affective similarity and relationship quality might be that 'affect similarity' is not a single, uniform concept in the literature. Researchers variously discuss emotional similarity, emotional convergence, co-experienced affect, shared emotions, synchrony, or positivity resonance, each carrying different theoretical assumptions (e.g., Anderson et al., 2003; Brown et al., 2022; Hinsz & Bui, 2023). Some work further frames affect in a core affect space defined by valence and arousal (Ekkekakis, 2013; Russell, 2003), while other work uses positive and negative affect composites, bipolar valence approaches, or discrete emotions (e.g., Brown et al., 2022; Kotiuga et al., 2025; Watson et al., 1988). These choices are not cosmetic. In line with the theoretical framing above, the present research explicitly distinguishes PA similarity (believed to be related to positive relationship outcomes) from NA similarity (believed to be related to negative relationship outcomes).

A second possible explanation is that theory and empirical research increasingly suggest that it is not affective similarity by itself, but the perception of such similarity that drives relationship outcomes (Montoya et al., 2008; Sels et al., 2020). What partners believe about similarity can be as important as (or more important than) what is objectively present. This suggests a useful reframing: rather than treating actual affect similarity as a direct interpersonal ingredient, it may often operate through interpretive processes (attention, inference, projection, empathic accuracy) that shape whether similarity becomes psychologically meaningful in the relationship. Since much of the existing research has measured either actual or perceived affect similarity in isolation instead of jointly, the relative importance of each remains unclear (From et al., 2025). Consequently, the present research models both actual similarity (what is jointly present in partners' reported experience) and perceived similarity (whether that sharedness is present from each partner's perspective) to disentangle these effects.

A final major reason the literature looks inconsistent is that studies quantify 'similarity' in very different ways. Classic measurement work already posited that similarity indices can combine multiple components: elevation (mean level), scatter/mismatch (dispersion), and shape (profile pattern) and that different operationalizations implicitly weight these components differently (Carlier et al., 2024; Cronbach & Gleser, 1953). Contemporary discussions extend this point in the dyadic similarity context and show that common indices (e.g., Euclidean distance, difference measures, Pearson correlation) can behave quite differently and may capture different information than researchers intend (Carlier et al., 2024; Furr, 2010; Kenny et al., 2020). For example, a pure distance score emphasizes mismatch but is intentionally blind to whether both partners are jointly high versus jointly low, correlation emphasizes shape covariation but can ignore elevation, and congruence or interaction-based approaches can become hard to interpret without careful probing (Edwards, 2002; Nestler et al., 2019). This matters directly for affect similarity because many relationship theories are not about 'closeness in scores', but about 'shared' affective experiences of sufficient intensity (especially for positive emotions) or shared negative affect under supportive conditions (Bodenmann, 2005; Boiger & Mesquita, 2012; Reis & Gable, 2015).

Recent research on ‘shared emotions’ nudges the field toward more precise conceptual targets. Brown et al. (2022), for instance, operationalized co-experienced affect as the temporal overlap of elevated affect and found that shared positive affect was linked to higher relationship quality, whereas shared negative affect was linked to lower relationship quality across conversation contexts. Theoretical work on positivity resonance similarly emphasizes that joint positive emotion can build connection and relational resources (Fredrickson, 2016). Taken together, these perspectives point to a key nuance: what may matter most is not generic similarity (being ‘close’ regardless of level), but jointly elevated affect with minimal mismatch, i.e., the extent to which partners are ‘up together’ (or ‘down together’) rather than merely ‘aligned somewhere’.

This is the conceptual move the present research also follows: (positive or negative) affect similarity is treated as shared, co-experienced elevated (positive or negative) affect, defined as joint elevation (in PA or NA) relative to each person’s baseline, coupled with low mismatch. The metric we use follows directly from this and from classic insights about the similarity components: we combine dyadic elevation (mean level) with a penalty for mismatch (absolute distance). Concretely, for a given moment we quantify similarity as the average of partners’ baseline-centered affect ratings minus the absolute difference between them (with parallel definitions for perceived similarity, comparing an actor’s affect to that actor’s perception of the partner’s affect). This mean intensity corrected by distance logic is common in other psychological contexts where co-occurrence and balance both matter, such as objective ambivalence indices (e.g., Schneider et al., 2015; Thompson et al., 1995). It also resonates with conjunctive or ‘weakest link’ perspectives in group functioning, where outcomes are constrained by the lowest input (Kozlowski & Ilgen, 2006; Steiner, 1972), and with related formalisms in fuzzy-set theory and profile overlap approaches (Smithson & Verkuilen, 2006).

Our operationalization does two things the broader literature has sometimes mixed up. First, it preserves the distinction between shared low and shared high: when partners are perfectly aligned, the score is larger when both partners are above baseline than when both are below baseline. Second, it treats mismatch as costly: even if both partners are relatively high, large divergences still reduce the degree to which the moment is considered a truly shared affective state. This is important because divergence can plausibly reflect missed attunement, regulatory burden, or interpersonal friction, processes emphasized in interpersonal affect dynamics (Butler, 2015).

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Subject*

Social and Behavioral Sciences: Psychology: Personality and Social Contexts

Social and Behavioral Sciences: Psychology: Quantitative Psychology

Tags

romantic couples; close relationships; dyadic processes; shared affect; co-experienced affect; affect similarity; perceived similarity; empathic accuracy; positivity resonance; we-ness; partner responsiveness; intensive longitudinal data; experience sampling; multilevel modeling; multilevel mediation.

Study Information

Hypotheses*

We study the same sample of romantic couples participated in two complementary study components, an ESM component and a laboratory interaction component. We refer to the ESM component as component 1 and the lab interaction component as component 2. Component 1 uses multilevel models of momentary ESM reports while component 2 uses lab models at the level of structured positive and negative lab interactions, rated via video-mediated recall.

Hypotheses for study component 1

We conceptualize affect similarity at the momentary level as the degree of shared or co-experienced elevated (positive or negative) affect, operationalized as the mean of the partners' baseline-centered ratings penalized for the magnitude of mismatch between them. We focus on similarity in both positive affect (PA) and negative affect (NA), and we distinguish between (a) actual similarity (actor's self-reported affect compared to partner's self-reported affect) and (b) perceived similarity (actor's self-reported affect compared to actor's perception of the partner's affect).

H1: Main effects of actual similarity on momentary love

We expect that being on the same affective wavelength in daily life relates to higher momentary love. We formulate separate hypotheses for PA and NA similarity:

- **H1.1 (actual PA similarity).** For PA, higher actual affect similarity between partners is associated with higher momentary love felt for the partner, both within dyads (moments when partners are more similar than usual for that couple) and between dyads (couples who are on average more similar than others).
- **H1.2 (actual NA similarity).** For NA, we predict the inverse pattern: higher actual NA similarity is associated with lower momentary love felt for the partner, both within and between dyads. When both partners feel similarly bad, this is expected to be negatively associated with love.

- **H1.3 (perceived similarity).** When perceived similarity is added to the models predicting momentary love (additively, without an interaction term), we expect perceived similarity to show associations with momentary love that are comparable in direction to those observed for actual similarity in models without perceived similarity. At the same time, we expect the association between actual similarity and momentary love to be reduced in magnitude when perceived similarity is included, consistent with the idea that perceived similarity partly accounts for (i.e., mediates) the association between actual similarity and love.

H2: Mediation by perceived similarity

The expectations in H1.3 are consistent with mediation, and we will therefore rely on a formal mediation model to test the proposed mechanism. Specifically, we will test whether (1) actual similarity predicts perceived similarity, (2) perceived similarity predicts momentary love when controlling for actual similarity, and (3) actual similarity remains associated with momentary love when controlling for perceived similarity. Together, these tests quantify the indirect association from actual similarity to love via perceived similarity, as well as any remaining direct association from actual similarity to love.

- **H2.1 (PA mediation).** The association between actual PA similarity and love is at least partially mediated by perceived similarity. Higher actual PA similarity is expected to predict higher perceived PA similarity, which in turn predicts higher love. This implies a positive indirect effect of actual PA similarity on love via perceived PA similarity.
- **H2.2 (NA mediation).** The association between actual NA similarity and love is at least partially mediated by perceived similarity. Higher actual NA similarity is expected to predict higher perceived NA similarity, which in turn predicts lower love. This implies a negative indirect effect.

While love may also shape perceived similarity (e.g., via projection), in the present study we focus on the pathway from actual similarity to perceived similarity to love, without attempting to establish temporal causality.

H3: Moderation by relational and contextual factors

We expect momentary perceived partner responsiveness and event valence to shape when similarity is helpful versus harmful, by testing whether the similarity–love associations depend on (a) the extent to which the actor feels validated and understood by the partner, and (b) the situational valence of the context (positive vs. negative moments).

When perceived responsiveness is high, affect similarity may be experienced as shared engagement and feeling understood, while when responsiveness is low, similarity may be experienced as mutual disengagement. In this way, responsiveness is expected to influence whether similarity functions as a relational resource or a relational burden:

- **H3.1 (actual/perceived PA similarity x responsiveness).** Higher momentary perceived partner responsiveness is expected to amplify the positive association between PA similarity (both actual and perceived) and love. When shared positivity occurs in the context of feeling understood and supported, affective alignment (or its

perception) should be especially likely to translate into higher love. In contrast, low responsiveness is expected to dampen the positive association between PA similarity and love.

- **H3.2 (actual/perceived NA similarity x responsiveness).** Higher momentary perceived responsiveness is expected to attenuate the negative association between NA similarity (both actual and perceived) and love. When both partners feel similarly bad but the actor experiences the partner as responsive and supportive, shared negativity may be interpreted as feeling understood and interpreted as 'we are in this together and coping jointly', such that the negative association between NA similarity and love is weakened. Inversely, low responsiveness is expected to strengthen the negative association between NA similarity and love.

In addition, similarity that does not 'fit' the moment may be more salient (e.g., surprising or discordant) and therefore more psychologically impactful. We expect therefore PA similarity to be more beneficial in relatively negative moments and NA similarity to be more harmful for love in relatively positive moments.

- **H3.3 (actual/perceived PA similarity x event valence).** We predict a mismatch effect: PA similarity will be most beneficial in negative moments. The positive association between PA similarity (both actual and perceived) and love will be stronger when the event/context is more negative than when it is more positive. Sharing PA in a negative or stressful context may be particularly salient and interpreted as joint resilience ('we can still enjoy this together'), thereby boosting momentary feelings of love more than shared positivity in already positive contexts.
- **H3.4 (actual/perceived NA similarity x event valence).** We predict a mismatch effect: NA similarity will be most detrimental in positive moments. The negative association between NA similarity (both actual and perceived) and momentary love will be stronger when the event/context is more positive than when it is more negative. When both partners feel similarly bad in an otherwise positive situation, this shared negativity may be especially discordant and thus more strongly associated with reduced feelings of love, compared to moments that are already negative.

Hypotheses for study component 2

Whereas component 1 focuses on how momentary affect similarity relates to momentary love in daily life, component 2 asks whether couples' average affect similarity during structured positive and negative lab interactions relates to post-interaction love/closeness. We again distinguish between actual and perceived similarity, and we conduct separate analyses for PA and NA similarity.

H1: Main effects of actual similarity on post-interaction relationship evaluations

We expect that being affectively on the same wavelength, on average, during a lab interaction relates to more positive post-interaction relationship evaluations. We formulate the following hypotheses to hold for both the negative and the positive lab interactions separately:

- **H1.1 (actual PA similarity).** Higher average actual PA similarity during the interaction predicts higher post-interaction love and closeness.
- **H1.2 (actual NA similarity).** Higher average actual NA similarity during the interaction predicts lower post-interaction love and closeness.
- **H1.3 (perceived similarity).** When interaction-averaged perceived similarity is added to the models (additively, without an interaction term), we expect perceived similarity to show associations in the same direction as those observed for actual similarity in models without perceived similarity. At the same time, we expect the association between actual similarity and post-interaction evaluations to be reduced in magnitude when perceived similarity is included, consistent with the idea that perceived similarity partly accounts for (i.e., mediates) the association between actual similarity and outcomes.

H2: Mediation by perceived similarity at the interaction level

The expectations in H1.3 are consistent with mediation, and we therefore perform a formal mediation analysis to test the proposed mechanism. We believe that this mechanism is general and therefore predict mediation in both the negative and positive interaction contexts:

- **H2.1 (PA mediation).** The association between interaction-averaged actual PA similarity and post-interaction love and closeness is at least partially mediated by interaction-averaged perceived PA similarity. Higher average actual PA similarity during the interaction is expected to predict higher average perceived PA similarity, which in turn predicts higher post-interaction love and closeness.
- **H2.2 (NA mediation).** The association between interaction-averaged actual NA similarity and post-interaction love and closeness is at least partially mediated by interaction-averaged perceived NA similarity. Higher average actual NA similarity during the interaction is expected to predict higher average perceived NA similarity, which in turn predicts lower post-interaction love and closeness.

H3: Moderation by relational and contextual factors

We expect (a) trait we-ness, defined as a stable relationship resource indicating the extent to which partners characteristically approach stressors and everyday challenges as a shared ‘we’, and (b) the interaction contextual valence (positive vs negative lab interaction) to moderate the relationship between similarity (both actual and perceived) and post-interaction love and closeness. First, we expect trait we-ness to function as an amplifier (for PA) or a buffer (for NA) across affective contexts, and thus test these hypotheses separately for both positive and negative interactions:

- **H3.1 (actual/perceived PA similarity x trait we-ness).** Higher trait we-ness is expected to amplify the positive association between interaction-averaged PA similarity (both actual and perceived) and post-interaction love and connectedness (i.e., PA similarity is more beneficial at higher trait we-ness). When couples high in we-ness share on average PA during the interaction (or perceive to do so), this should be

particularly linked to higher post-interaction love and closeness, compared to couples low in we-ness.

- **H3.2 (actual/perceived NA similarity x trait we-ness).** Higher trait we-ness is expected to attenuate the negative association between interaction-averaged NA similarity (both actual and perceived) and post-interaction love and closeness (i.e., NA similarity is less harmful at higher trait we-ness). When both partners feel on average similarly bad during an interaction (or perceive to do so) but experience high we-ness, this can be interpreted as 'we are in this together and coping jointly', such that the negative association between NA similarity and post-interaction evaluations is less negative at higher trait we-ness.

Second, we predict that the valence of the interaction itself determines whether similarity is helpful or harmful.

- **H3.3 (actual/perceived PA similarity x interaction valence).** We predict a mismatch effect: PA similarity will be most beneficial during negative interactions. The positive association between interaction-averaged PA similarity (actual and perceived) and post-interaction love and connectedness is expected to be stronger (more positive) in the negative interaction than in the positive interaction. Sharing PA on average (or perceiving to do so) in a negative or stressful interaction may be particularly salient and interpreted as 'we can still enjoy this together', boosting post-interaction love and closeness more than shared positivity in already positive interactions.
- **H3.4 (actual/perceived NA similarity x interaction valence).** We predict a mismatch effect: NA similarity will be most detrimental during positive interactions (where it violates the context's valence). The negative association between interaction-level NA similarity (actual and perceived) and post-interaction love and connectedness is expected to be stronger (more negative) in the positive interaction than in the negative interaction. When both partners feel similarly bad (or perceive to do so) in an otherwise positive interaction, this shared negativity may be especially discordant and thus more strongly associated with reduced post-interaction love and closeness.

Design Plan

Study type*

Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, "natural experiments," and regression discontinuity designs.

Blinding*

No blinding is involved in this study.

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Is there any additional blinding in this study?

N/A

Study design*

104 couples (i.e., 208 individuals) were recruited via psychiatry and psychotherapy clinics and flyers distributed through social media and public/clinical settings. Inclusion criteria were: being over the age of 18, being in a relationship for at least two months, living together, not residing in a treatment facility, being fluent in Dutch, and both partners owning a smartphone.

The full procedure of the project included filling out a SCID screener, a SCID interview (if the participant reported elevated scores on major depressive disorder, bipolar disorder, or borderline personality disorder or if the participant fit to an exclusion criteria, namely lifetime psychotic disorder or current substance use disorder), completing the background questionnaires, and having a positive and negative lab interaction, in which partners discussed a topic that they were satisfied with and disagreed about in their relationship. Next, both members of the couples completed the experience sampling (ESM) procedure, where they responded to short questionnaires sent to their smartphones for two weeks. In the present study, we will use the data from the background questionnaires, negative and positive lab interactions, and the ESM procedure.

Experience Sampling Procedure

The ESM period lasted for 14 days. Partners received notifications (beeps) to complete questionnaires on their smartphones. They received the beeps 5 times on weekdays between 5 pm and 10.15 pm, and 10 times on weekend days between 10:45 am and 10:15 pm. This was done to maximize the odds of couples receiving beeps when they are together. The interval between the starting and the end times of sending the beeps was divided into equal 1h intervals, and beeps were sent at a random time in those intervals to both partners at the same time. We ensured that there were at least 15 minutes between the consecutive beeps. At each beep, partners answered questions about their feelings, behaviors toward their partners, as well as their perception of their partner's feelings and behaviors. The questionnaire consisted of 26-28 questions (with conditional branching) and lasted approximately 2 minutes. If participants didn't respond to a beep within one minute, they received a reminder. If they didn't respond within five minutes, the beep was no longer available, meaning that their data was considered missing for that beep. The study protocol was approved by the ethics committee, and all participants gave written informed consent. During the study, the monetary compensation was adjusted from a maximum of €70 per person to a maximum of €100 per person, depending on the compliance.

Lab interaction

For the lab interaction, the procedure included engaging in a positive and separately in a negative interaction. Couples were first asked to discuss events of the day for 2 minutes to get comfortable with interacting with each other in the lab setting. Next, they discussed a topic of

disagreement for 8 minutes. The topics for the positive and negative discussions in the interaction was determined based on both members' scores from the Problem Areas Questionnaire (PAQ; Heavey et al., 1995). The PAQ consists of potential themes of issues and strengths in romantic relationships (e.g., finances, children, views on life). For the negative interaction, a topic was selected if both partners had scored the area as problematic or if their ratings showed a large discrepancy. For the positive interaction, the topic with the highest summed scores across both partners was used. When multiple topics were equally negative or positive, one was chosen at random. This procedure ensured that topics differed across couples, but always included one area perceived negatively by at least one partner and one area perceived positively by both. If no negative topics could be selected, participants were instructed to think about a negative topic to discuss.

After each interaction, each partner completed a short questionnaire in a private cubicle about how they had felt and experienced the discussion.

Three cameras recorded the interaction: two facing each partner separately, one recording the overview of the interaction. After the interactions, a Video Mediated Recall (VMR) procedure was used. Both members of the couples watched the recording of the interaction from their perspective (i.e., the recording from the camera facing their partners). The recording was paused every 30 seconds and participants were asked to report on their own and partner's feelings and interpersonal behaviors at that 30-second interval.

Randomization

There was no randomization in the study.

Sampling Plan

Existing data*

Registration prior to accessing the data: As of the date of submission, the data exist, but have not been accessed by you or your collaborators. Commonly, this includes data that has been collected by another researcher or institution.

Explanation of existing data

The data was collected for a larger project as part of the RE-Connect Consortium. The present study will use the background questionnaires, the video mediated recall data from the negative lab interaction, the post interaction questionnaires, and the ESM data. The data codebooks were accessed (to know what data was available). The first author has not accessed any data. The second author has seen data related to other non-related research questions. Overall, no data relevant to the present hypotheses were inspected prior to preregistration.

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Data collection procedures*

Couples were recruited via psychiatry and psychotherapy clinics and flyers distributed through social media and public/clinical settings in Belgium and the Netherlands. Inclusion criteria were: being over the age of 18, being in a relationship for at least two months, living together, not residing in a treatment facility, being fluent in Dutch, and both partners owning a smartphone. During the study, the monetary compensation was adjusted from a maximum of €70 per person to a maximum of €100 per person, depending on the compliance. The study started on 19/04/2023 and the ESM procedure was completed for all participants on 11/04/2025.

Sample size*

104 couples were recruited (i.e., 208 individuals) for the larger project. However, we will exclude one couple, as technical issues made it unclear which ESM questionnaire responses correspond to which partner. Thus, the data from 103 couples will be analyzed for the analysis of the ESM data. Moreover, 6 couples (including the couple whose ESM data we will exclude) did not complete the lab session. Thus, we will analyze the data of 98 couples for the analysis on lab interaction data.

Sample size rationale

The available sample will be used for the present study. Models will be specified parsimoniously, to avoid oversaturation and preserve power. Sensitivity and robustness checks (e.g., with extended models) will be performed to further validate the models.

Stopping rule

Based on power analysis that was performed for the initial Re-Connect data collection, the target sample size was 100 couples (i.e., 200 individuals). After starting the data collection, additional couples were recruited to account for potential exclusions due to unreliable data from some enrolled participants – totaling to 104 couples (i.e., 208 individuals).

Variables

Manipulated variables

N/A

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Measured variables *

Study component 1: Experience Sampling data
Participants answered a short questionnaire (±2 minutes) through momentary notifications sent to their smartphones every day for two weeks.

Outcome variables (time-variant)

- **Loving towards partner:** Participants were asked “How loving do you feel towards your partner at the moment?” and answered using a 0-100 slider scale, ranging from 0 = “not at all loving” to 100 = “very loving”.

Predictor variables (time-variant)
Similarity constructs based on

- **Negative affect:** Participants were asked “How negative do you feel at this moment?” and answered using a 0-100 slider scale, ranging from 0 = “not negative at all” to 100 = “very negative”.
- **Positive affect:** Participants were asked “How positive do you feel at this moment?” and answered using a 0-100 slider scale, ranging from 0 = “not positive at all” to 100 = “very positive”.
- **Negative affect partner:** Participants were asked “How negative do you think your partner feels at this moment?” and answered using a 0-100 slider scale, ranging from 0 = “not negative at all” to 100 = “very negative”.
- **Positive affect partner:** Participants were asked “How positive do you think your partner feels at this moment?” and answered using a 0-100 slider scale, ranging from 0 = “not positive at all” to 100 = “very positive”.

Used for moderation hypotheses (time-variant)

- **Perceived partner responsiveness:** Participants were asked “To what extent do you feel understood by your partner at this moment?” and answered using a 0-100 slider scale, ranging from 0 = “not at all” to 100 = “completely”.
- **Negative event general:** Participants were asked “Did something negative happen since the last beep?”, using a multiple choice response: 1 = ”No”, 2= ”Yes, a little”, 3= “Yes, a lot”.
- **Positive event general:** Participants were asked “Did something positive happen since the last beep?”, using a multiple choice response: 1 = ”No”, 2 = ”Yes, a little”, 3 = “Yes, a lot”.

Study component 2: Lab interactions data and background questionnaire

Outcome variables (time-invariant; from post-interaction questionnaires)

- **Loving:** Participants were asked “How loving do you feel towards your partner at the moment?” and answered using a 0-100 slider scale, ranging from 0 = ”not at all” to 100 = ”very”.

- **Closeness:** Participants were asked “How close do you feel towards your partner at the moment?” and answered using a 0-100 slider scale, ranging from 0 = ”not at all” to 100 = ”very”.

Predictor variables (time-variant from the VMR, but used for similarity construct, averaged over the interaction)

- **Own negative affect:** Participants were presented with the statement “During the part of the conversation I just saw, I felt negative”, answering using a 0-100 slider scale with 0 = ”not negative at all” and 100 = ”very negative”.
- **Own positive affect:** Participants were presented with the statement “During the part of the conversation I just saw, I felt positive”, answering using a 0-100 slider scale with 0 = ”not positive at all” and 100 = ”very positive”.
- **Partner negative affect:** Participants were presented with the statement “During the part of the conversation I just saw, I think my partner was feeling negative”, answering using a 0-100 slider scale with 0 = ”not negative at all” and 100 = ”very negative”.
- **Partner positive affect:** Participants were presented with the statement “During the part of the conversation I just saw, I think my partner was feeling positive”, answering using a 0-100 slider scale with 0 = ”not positive at all” and 100 = ”very positive”.

Background questionnaire

- **Dyadic Coping Inventory:** Participants completed the Dutch version of the Dyadic Coping Inventory (DCI; Bodenmann et al., 2018; Ponnet, 2012). The scale consists of 37 items assessing dyadic coping, that is, how partners communicate about stress and how they support (or hinder) each other’s coping, including supportive, delegated, negative, and common/joint coping responses (e.g., “I show my partner through my behavior when I am not doing well or when I have problems”, “My partner shows empathy and understanding”). However, items 31 to 37 (covering common/joint coping and overall evaluation) were not implemented in Qualtrics and were therefore missing. The answers were on a scale of 1 to 5 (i.e., 1 = “very rarely” to 5 = “very often”).
- The items from the DCI are used to construct trait we-ness. Because only 30 DCI items are available in this study, we cannot include common dyadic coping (items 31–35) or evaluation items (36–37). For each person, a trait we-ness score will therefore be calculated as the mean of the 22 items reflecting constructive stress communication and supportive dyadic coping by self and partner (supportive and delegated):
    1. **Stress Communication (Self & Partner):**
       Items 1, 2, 3, 4, 16, 17, 18, and 19, reflecting the open expression of stress and feelings.
    2. **Supportive Dyadic Coping (Perceived Partner Support):**
       Items 5, 6, 8, 9, 12, 13, and 14, reflecting emotion-focused and problem-focused support received from the partner.

3. **Supportive Dyadic Coping (Own Provided Support):** Items 20, 21, 23, 24, 28, 29, and 30, reflecting emotion-focused and problem-focused support provided to the partner.

We will exclude items 7, 10, 11, 15, 22, 25, 26, and 27, related to negative dyadic coping (hostile, ambivalent, or superficial coping) to ensure the score represents positive 'we-ness'.

Indices

We build on psychological measurement approaches that combine joint intensity with a penalty for imbalance. In objective ambivalence literature, the objective ambivalence index between P(ositive) and N(egative) is calculated as a function of mean intensity corrected by distance: (Schneider et al., 2015; Thompson et al., 1995). Concretely, we construct a similarity index as the difference between (a) dyadic elevation (the momentary mean level of the two partners) and (b) a penalty for mismatch (the absolute distance between two affect levels).

For study component 2, we average the similarity indices calculated from the Video Mediated Recall data over the interaction.

Analysis Plan

Statistical models *

For Study Component 1 (on ESM data)

General Model Specifications

Positive and negative affect. We will perform separate analyses for positive affect (PA) and negative affect (NA). In what follows, we use 'affect' as an umbrella term for PA and NA and analyses are conducted separately for each.

Dyad indistinguishability. Because our data may include dyads where partners share the same gender, and because our research questions are symmetric for both partners, we will treat dyads as indistinguishable. We will not model specific roles but estimate a single set of effects for both partners, which simplifies the model structure and helps increase parsimony and statistical power.

Notation and centering. There are three levels in the data with moments (m) nested within persons (p) nested within dyads (d). This multilevel structure follows standard practice in experience sampling research with couples (Bolger & Laurenceau, 2013; Kenny et al., 2020). Let L_{mpd} denote momentary love felt for the partner by person p in dyad d at moment m. Let A_{mpd} denote the actor's own (self-reported) momentary affect. For the partner's affect, we

distinguish between p's partner's actual (self-reported) momentary affect, P_{mpd}^{act}, and the partner affect as perceived by the actor, P_{mpd}^{perc}. Note that if p' denotes person p's partner, $P_{mpd}^{\text{act}} \equiv A_{mp'd}$. A_{mpd}, P_{mpd}^{act} and P_{mpd}^{perc} are all measured on the same 0-100 scale and are thus commensurable (Edwards, 2002; Humberg et al., 2019).
We will person-mean center the momentary affect variables to focus on within-person fluctuations and to reduce multicollinearity between predictors and their interaction terms, which also improves the interpretability of the main effect predictors (Aiken & West, 1991; Cohen et al., 2003; Enders & Tofighi, 2007). Therefore, we define person specific means across all moments as

$$\bar{A}_{pd} = \frac{1}{M_{pd}} \sum_{m=1}^{M_{pd}} A_{mpd}\,, \quad \bar{P}_{pd}^{act} = \frac{1}{M_{p'd}} \sum_{m=1}^{M_{p'd}} P_{mpd}^{\text{act}}\,, \quad \bar{P}_{pd}^{perc} = \frac{1}{M_{pd}} \sum_{m=1}^{M_{pd}} P_{mpd}^{\text{perc}},$$

With person p' denoting person p's partner and M_{pd} ($M_{p'd}$) the number of observed moments for person p (p') in dyad d. We then define person-mean centered predictors as

$$X_{mpd} = A_{mpd} - \bar{A}_{pd},\ Y_{mpd}^{act} = P_{mpd}^{\text{act}} - \bar{P}_{pd}^{act}\,, \text{and } Y_{mpd}^{perc} = P_{mpd}^{\text{perc}} - \bar{P}_{pd}^{perc}.$$

Thus, X_{mpd} denotes the person-mean centered actor affect, Y_{mpd}^{act} the corresponding person-mean centered actual partner affect, and Y_{mpd}^{perc} the mean centered perceived partner affect. Remark that we will not z-standardize the momentary affect variables (globally or within person), because this could distort individual differences and between-person relations and is not recommended for longitudinal multilevel models (Moeller, 2015). Moreover, keeping all affect variables on the original 0-100 scale, preserves a straightforward interpretation of a given raw difference.

Definition of similarity indices. The multidimensional, seminal framework established by Cronbach and Gleser (1953) posits that 'similarity' is not a unitary construct but can reflect different components: *elevation* (mean level), *scatter* (dispersion or mismatch), and *shape* (the pattern of peaks and valleys across items). They highlighted that similarity indices can combine these three components in different ways, and that different operationalizations implicitly weight components differently. Also more recent discussions point out that similarity indices may reflect mixtures of elevation, scatter, and shape, and that these components are statistically intertwined in practice (Carlier et al., 2024; Furr, 2010; Kenny et al., 2020).

In our research questions, the conceptual focus is not on generic affect similarity (i.e., being 'close', regardless of whether both partners are high vs low), but on shared or co-experienced elevated (positive or negative) affect. While shared affect has previously been measured as the temporal overlap of elevated emotions (e.g., Brown et al., 2022), our focus on momentary dynamics requires a metric that quantifies this sharedness between single affect ratings rather than aggregating duration over a period of time. Specifically, we target joint elevation in (positive or negative) affect, relative to each person's baseline, coupled with low mismatch. A pure distance-based operationalization (e.g., as in Sels et al., 2020) primarily captures discrepancy / mismatch and is intentionally blind to whether partners are jointly high vs jointly low relative to baseline (i.e., it omits elevation). This is problematic for our conceptual target,

because (after person-mean centering) 'high similarity' should not treat (−50, −50) as equivalent to (+50, +50).

Beyond this conceptual mismatch, purely difference-based indices can also be psychometrically weaker when the two components are positively correlated (as partners' affect often is): for the algebraic difference, $\mathrm{Var}(X-Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)-2\mathrm{Cov}(X,Y)$, so stronger covariation shrinks the true-score variance, while measurement error from both components accumulates, lowering reliability (Edwards, 2002).

In the present research we therefore build on psychological measurement approaches that combine joint intensity with a penalty for imbalance. In objective ambivalence literature, the objective ambivalence index between P(ositive) and N(egative) is calculated as a function of mean intensity corrected by distance: $(P+N)/2-|P-N|$ (Schneider et al., 2015; Thompson et al., 1995). Concretely, we operationalize similarity as a composite of (a) dyadic elevation (the momentary mean level of the two partners) and (b) a penalty for mismatch (the absolute distance).

Such a composite also has modelling advantages. Our similarity quantification retains an explicit shared-elevation component alongside a mismatch penalty. As a result, similarity is likely less dominated by the reliability limitations of pure difference scores and may yield more stable estimates in dyadic affect data (Edwards, 2002).

We define for actual affect

$$M_{mpd}^{\mathrm{act}}=\frac{X_{mpd}+Y_{mpd}^{\mathrm{act}}}{2},\ \ D_{mpd}^{\mathrm{act}}=|X_{mpd}-Y_{mpd}^{\mathrm{act}}|.$$

We then define actual affect similarity as:

$$S_{mpd}^{\mathrm{act}}=M_{mpd}^{\mathrm{act}}-D_{mpd}^{\mathrm{act}}.$$

Lower values reflect either jointly low (below-baseline) affect and/or mismatch while higher values reflect jointly elevated, aligned affect.

Analogously, for perceived affect we define:

$$M_{mpd}^{\mathrm{perc}}=\frac{X_{mpd}+Y_{mpd}^{\mathrm{perc}}}{2},\quad D_{mpd}^{\mathrm{perc}}=|X_{mpd}-Y_{mpd}^{\mathrm{perc}}|,\ \ S_{mpd}^{\mathrm{perc}}=M_{mpd}^{\mathrm{perc}}-D_{mpd}^{\mathrm{perc}}.$$

This operationalization matches our conceptual target. First, it distinguishes shared low from shared high: if for example partners are perfectly aligned ($X_{mpd}=Y_{mpd}^{\mathrm{act}}$), then $S_{mpd}^{\mathrm{act}}=X_{mpd}$, so $(-50,-50)$ and $(+50,+50)$ are not treated as equally 'high similarity'. Second, it explicitly penalizes mismatch, ensuring that extreme divergences are scored as less similar than mild divergences, consistent with the idea that larger divergence may reflect greater relationship impacts.

Actual affect similarity is symmetric within a couple at a given moment (it compares the two partners' self-reported affect), such that S_{mpd}^{act} is conceptually a dyad–moment construct that is identical for both partners. It is defined only for moments in which both partners provided self-

reports. In contrast, perceived similarity is actor-specific because it compares the actor's affect to the actor's perception of the partner's affect, so S_{mpd}^{perc} can differ between partners within the same dyad and is defined for moments in which the actor provided a report.

Note that our operationalization also aligns with conjunctive or 'weakest link' models of group processes (Steiner, 1972; see also Kozlowski & Ilgen, 2006), where a dyadic outcome is determined by the member with the lower functional state. In the domain of fuzzy set theory and profile similarity, this logic is formalized as the intersection or overlap of two values, operationalized as the minimum score (Carlier et al., 2024; Smithson & Verkuilen, 2006). Mathematically, this standard minimum is equivalent to subtracting only half the mismatch from the mean level (M-0.5D). Our measure (M-D) essentially adopts this conjunctive foundation but extends it by imposing a stricter penalty for mismatch. This reflects the theoretical position that large affective divergences are not merely 'non-events' but active sources of regulatory effort and instability, warranting a score lower than the simple minimum of the two partners.

Decomposition of similarity indices. Because similarity varies moment-to-moment and also differs on average across dyads and persons, we decompose the similarity indices into within- and between-components, following recommendations for separating within from between effects in multilevel models (Bolger & Laurenceau, 2013; Enders & Tofighi, 2007; Kenny et al., 2020). For actual similarity (which is symmetric within dyads at each moment) these are within-dyad and between-dyad, whereas for perceived similarity (which is actor-specific) these are within-person and between-person components.

For each dyad d, we compute the dyad-mean actual similarity across all observed moments and both partners:

$$\bar{S}_d^{act} = \frac{1}{N_d^{act}} \sum_{p,m} S_{mpd}^{act}$$

where N_d^{act} denotes the number of person-moments (across partners) for which S_{mpd}^{act} is defined in dyad d. Because S_{mpd}^{act} is symmetric, averaging across person-moments is equivalent to averaging across dyad-moments. Let $\bar{\bar{S}}^{act}$ denote the grand mean across dyads. We then define:

$$S_{mpd}^{act,\,\text{within}} = S_{mpd}^{act} - \bar{S}_d^{act}, \quad \bar{S}_d^{act,between} = \bar{S}_d^{act} - \bar{\bar{S}}^{act}.$$

For each person p in dyad d, we compute the person-mean perceived similarity across moments:

$$\bar{S}_{pd}^{perc} = \frac{1}{N_{pd}^{perc}} \sum_{m} S_{mpd}^{perc}$$

where N_{pd}^{perc} denotes the number of moments for which S_{mpd}^{perc} is defined for that person. Let $\bar{\bar{S}}^{perc}$ denote the grand mean across persons and dyads. We then define

$$S_{mpd}^{perc,\,\text{within}} = S_{mpd}^{perc} - \bar{S}_{pd}^{perc}, \quad \bar{S}_{pd}^{perc,between} = \bar{S}_{pd}^{perc} - \bar{\bar{S}}^{perc}.$$

The within components capture momentary deviations from a dyad's (for actual similarity) or a person's (for perceived similarity) typical similarity, whereas the between components capture stable differences between dyads (for actual similarity) or between persons (for perceived similarity). For interpretation, because persons are nested in dyads, the between-person perceived similarity component may combine between-dyad differences and partner differences within dyads.

Random intercepts. We will include normally distributed random intercepts for dyads and persons nested within dyads in all primary models. Consistent with recommendations for multilevel modelling (e.g., Hox et al., 2017; Nestler et al., 2019), we are mindful that a full random-effect structure can lead to convergence problems, particularly for models with many interaction terms and complex random-effect structures. To prioritize parsimony and reduce convergence problems in models with multiple interaction terms, we will not include random slopes in the primary models. Random slopes for linear terms will be examined as a separate robustness/sensitivity check (see below).

Momentary temporal dependence. To account for temporal dependence in the ESM reports, we will specify a first order autoregressive [AR(1)] covariance structure for the level 1 residuals within persons (nested in dyads). This means that residuals from nearby measurement moments for the same person in a dyad are allowed to correlate, with the correlation decreasing as the lag between observations increases. Because beeps are unequally spaced, we will implement AR(1) using the actual time stamps (continuous-time AR(1)) when supported by the software and the model converges. Otherwise we will use lag-by-beep AR(1) ($\mathrm{Corr}(\varepsilon_{mpd}, \varepsilon_{m+k,pd}) = \rho^k$, $|\rho| < 1$) as an approximation. Residuals from different persons are assumed independent. This is common practice in intensive longitudinal and affect dynamics research and helps to obtain appropriate standard errors for the fixed effects (Bolger & Laurenceau, 2013; Hamaker et al., 2015). If AR(1) structures lead to convergence problems or unacceptable parameter estimates (for example $|\rho| \geq 1$), we will revert to models with independent residuals and report this deviation.

Parsimony and robustness checks (sensitivity analyses). In principle, models can be sensitive to actor and partner affect levels that may correlate with the outcome. For example, our similarity measure will tend to be higher when both partners are simultaneously high relative to baseline. At the same time, momentary love may also be related to the individual affect levels. As a result, an observed similarity–love association could reflect similarity (as intended) or simply that moments with higher similarity are also moments where one or both partners feel better (or worse). To check that our similarity effects are not merely capturing these main effects of actor and partner affect, it is common to include actor/partner affect levels (i.e., elevation) as covariates (Edwards, 2002; Griffin et al., 1999). Additionally, slow shifts in couples' experiences, or gradual habituation to the study may lead to systematic trends over the course of the ESM study. To capture this, a time trend variable could be added to the models, for example as a linear time covariate or as separate day-to-day and within-day covariates (Bolger & Laurenceau, 2013; Hamaker et al., 2015). However, adding all these covariates alongside decomposed similarity indices can oversaturate the models and impair convergence. We therefore prioritize parsimonious primary models, and will evaluate

robustness via sensitivity checks. In addition, fixed-slope models assume that within-dyad similarity effects are the same for all persons/couples. As a robustness check, we will evaluate whether allowing these slopes to vary (random slopes) changes substantive conclusions.

Because combining all sensitivity and robustness tests in one model is likely to produce non-convergence, these checks will be run in separate model variants:

(1) **Elevation sensitivity.** We will additionally fit models including the individual (actual or perceived) affect levels as covariates to check whether the effect of similarity changes meaningfully. Specifically, when feasible (i.e., converging), we will add the person-mean centered actor and partner affects and the corresponding average actor and partner affect levels across the ESM period (grand-mean centered) as covariates. In other words, we test whether 'we feel the same' predicts love beyond simply 'I feel good'. We will also inspect correlations between momentary love and the candidate covariates. If these correlations are small (e.g., |r| < .30; following the reasoning in Sels et al., 2020), this provides some reassurance even if the larger sensitivity models do not converge.

(2) **Time-trend sensitivity.** Rather than including time covariates in all primary models, we will test for systematic time trends in momentary love across the ESM period (e.g., habituation or drift across the two-week period), for example by fitting $L_{mpd} = b_{0,pd} + \beta_T T_m + \varepsilon_{mpd}$, with T_m the chronological order of beeps, centered at the study midpoint. When indicated, we will re-estimate models including a linear time covariate (and, where feasible, day-in-study and time-of-day) as an additional robustness check. As an additional diagnostic, we will also inspect whether the similarity indices show systematic time trends.

(3) **Random-slope sensitivity.** Separately from the covariate and time-trend checks, we will attempt models that add random slopes for key level-1 predictors (e.g., within-dyad actual similarity and/or within-person perceived similarity).

Across sensitivity checks, we will interpret the similarity effects as robust if they remain comparable in direction and magnitude (not merely in statistical significance).

Step 1a: Multilevel decomposition models using actual similarity (H1.1–H1.2)

We first test whether momentary love is predicted by momentary actual affect similarity, decomposed into within- and between-dyad components. We estimate:

$$L_{mpd} = b_{0,pd} + \beta_1 S_{mpd}^{act,\,within} + \beta_2 \bar{S}_d^{act,\,between} + \varepsilon_{mpd},$$

with random intercept

$$\mathrm{b} = \beta_0 + v_{0,d} + u_{0,pd},$$

where β_0 is the fixed intercept, $v_{0,d} \sim N(0, \tau_d^2)$ is the dyad-level random intercept, $u_{0,pd} \sim N(0, \tau_p^2)$ is the person-level random intercept, and ε_{mpd} is the level 1 residual, modelled with an AR(1) covariance structure within persons as described above.

In this model, β_1 and β_2 quantify the within- and between-dyad associations between actual similarity and love. H1.1 predicts positive β_1 and β_2 for PA and H1.2 predicts negative β_1 and $\beta_2$for NA.

Step 1b: Adding perceived similarity (H1.3)

Next, we add perceived similarity alongside actual similarity (without an interaction term) to evaluate whether perceived similarity predicts love above and beyond actual similarity, and whether the actual similarity coefficients are reduced in magnitude when perceived similarity is included (H1.3).

$$L_{mpd} = b_{0,pd} + \beta_1 S_{mpd}^{act,\,within} + \beta_2 \bar{S}_d^{act,\,between} + \beta_3 S_{mpd}^{perc,\,within} + \beta_4 \bar{S}_{pd}^{perc,\,between} + \varepsilon_{mpd},$$

with the same random-intercept structure as in Step 1a.

We do not include the actual x perceived interaction in this step to keep the test of H1.3 parsimonious and directly interpretable: in an additive model, the perceived-similarity coefficients quantify their unique associations with love above and beyond actual similarity. We examine conditional (interaction) effects separately as an exploratory extension in Step 1c.

The coefficient β_3 indicates whether at moments when an actor perceives more similarity than they typically do (i.e., higher $S_{mpd}^{perc,within}$), they report higher/lower love over and above what can be explained by actual similarity at that moment.

The coefficient β_4 indicates whether persons who tend to perceive more similarity on average report higher/lower love overall, controlling for the dyad's average actual similarity. As noted for interpretation, this between-person component combines differences between dyads and stable differences between partners within dyads.

We will interpret evidence consistent with H1.3 as (a) β_3 and $\beta_4$being comparable in direction to the corresponding actual-similarity effects in Step 1a, and (b) the actual-similarity association with love being reduced in magnitude when perceived similarity is included, operationalized as attenuation of β_1 and/or β_2 relative to Step 1a.

Step 1c (exploratory): Joint effects, interaction, and probing

As an exploratory extension, we will add interaction terms between actual and perceived similarity to examine whether the association of each form of similarity with love depends on the level of the other. To align with the within–between decomposition, we include separate interaction terms for the within components (dyad-centered for actual similarity and person-centered for perceived similarity) and for the between components (dyad-level for actual similarity and person-level for perceived similarity):

$$\begin{aligned} L_{mpd} = {} & b_{0,pd} + \beta_1 S_{mpd}^{act,\,within} + \beta_2 \bar{S}_d^{act,\,between} + \beta_3 S_{mpd}^{perc,\,within} + \beta_4 \bar{S}_{pd}^{perc,\,between} \\ & + \beta_5 \left(S_{mpd}^{act,\,within} \times S_{mpd}^{perc,\,within}\right) + \beta_6 \left(\bar{S}_d^{act,\,between} \times \bar{S}_{pd}^{perc,\,between}\right) + \varepsilon_{mpd}, \end{aligned}$$

with the same random-intercept structure as in Step 1a.

For interpretation, we will probe predicted love for combinations of low (−1 SD), average, and high (+1 SD) values of actual and perceived similarity (see e.g., Aiken & West, 1991; Cohen et al., 2003):

- The within interaction coefficient β_5 indicates whether the momentary association between one form of similarity and love depends on the other form of similarity at that same moment. We will interpret β_5 by examining predicted love at combinations of low, average, and high values of $S_{mpd}^{act,within}$ and $S_{mpd}^{perc,within}$, with between components held at their means (i.e., 0 due to centering). These probes allow us to evaluate: (a) whether perceived similarity is associated with higher love when actual similarity is low (consistent with an 'illusory benefit'), and (b) whether actual similarity is most predictive when perceived similarity is high (i.e., when being similar also 'feels' similar).
- The between-component interaction coefficient β_6 indicates whether the association between dyads' average actual similarity and love depends on persons' typical perceived similarity, and vice versa. We will interpret β_6 by examining predicted love at combinations of low, average, and high values of $\bar{S}_d^{act,between}$ and $\bar{S}_{pd}^{perc,between}$, with within-person components held at their means (i.e., 0 due to centering). These probes allow us to evaluate: (a) whether persons' typical perceived similarity is associated with love even when dyads are low in average actual similarity, and (b) whether dyads' average actual similarity is most predictive of love when persons are high in typical perceived similarity.

These interaction tests are exploratory, and we do not formulate specific directional hypotheses for β_5 or β_6.

Step 2: Multilevel mediation (H2.1-H2.2)

To test the process hypothesis that momentary actual affect similarity predicts perceived affect similarity, which in turn predicts higher love felt (H2.1) or lower love felt (H2.2), we will conduct multilevel mediation analyses. We will also explore if actual similarity retains a direct unique association with love. These mediation models test an indirect association pattern consistent with the proposed process, but do not establish causal ordering.

The mediator model (perceived similarity) is:

$$S_{mpd}^{\text{perc}} = t_{0,pd} + \theta_1 S_{mpd}^{\text{act,within}} + \theta_2 \bar{S}_d^{\text{act,between}} + \varepsilon_{mpd}^{M},$$

with random intercept

$$t_{0,pd} = \theta_0 + v_{0,d}^{M} + u_{0,pd}^{M},$$

where $v_{0,d}^{M}$ and $u_{0,pd}^{M}$ are the dyad-level and person-level random intercepts, and ε_{mpd}^{M} is modelled with an AR(1) covariance structure within persons.

Here, S_{mpd}^{perc} is the momentary mediator. As before, we decompose actual similarity into a within-dyad (momentary deviation from dyad mean) component and a between-dyad mean

component. Therefore, θ_1 represents the momentary association between actual and perceived similarity, whereas θ_2 represents the between-dyad association.

The outcome model (love) is:

$$L_{mpd} = f_{0,pd} + \varphi_1 S_{mpd}^{\text{perc, within}} + \varphi_2 \bar{S}_{pd}^{\text{perc, between}} + \varphi_3 S_{mpd}^{\text{act, within}} + \varphi_4 \bar{S}_{d}^{\text{act, between}} + \varepsilon_{mpd}^{L},$$

with random intercept

$$f_{0,pd} = \varphi_0 + v_{0,d}^{L} + u_{0,pd}^{L},$$

where $v_{0,d}^{L}$ and $u_{0,pd}^{L}$ are the dyad-level and person-level random intercepts, and ε_{mpd}^{L} is modelled with an AR(1) covariance structure within persons.

The within-person (momentary) direct effect of actual similarity on love is quantified as φ_3, while the between-dyad direct effect is quantified as φ_4. The within-person (momentary) indirect effect of actual similarity on love via perceived similarity is quantified as $\theta_1\varphi_1$, where θ_1 is the within-person association between actual similarity and perceived similarity in the mediator model and φ_1 is the within-person association between perceived similarity and love in the outcome model (controlling for actual similarity). The between-component indirect effect is quantified by $\theta_2\varphi_2$ (dyad-level actual similarity → person-level typical perceived similarity → love). Note that although θ_2 pertains to a dyad-level predictor and φ_2 pertains to a between-person mediator component, their product captures a between-level indirect association: dyad-average actual similarity is associated with stable (intercept-level, it doesn't change moment-to-moment) differences in perceived similarity, which in turn are associated with love (Preacher et al., 2010; Zhang et al., 2009).

For the indirect effects, we will obtain confidence intervals using the Monte Carlo simulation method for multilevel mediation (e.g., Bauer et al., 2006; Preacher et al., 2010), as bootstrapping can be too computationally heavy in complex multilevel models.

Step 3a: Moderation by perceived partner responsiveness (H3.1–H3.2)

To test H3.1–H3.2, we examine whether the similarity–love association depends on momentary perceived partner responsiveness. Perceived partner responsiveness is measured on a 0–100 scale. For each person p in dyad d at moment m, let R_{mpd} denote momentary perceived partner responsiveness. Consistent with our general centering strategy, we person-mean center R_{mpd} across all moments of that person, so 0 corresponds to person p's typical level of perceived partner responsiveness. We focus on within-person moderation by person-mean centering R_{mpd}; we do not model between-person differences in typical responsiveness as a moderator in the primary models.

For actual similarity, the moderation model extends Step 1a.

We estimate:

$$L_{mpd} = b_{0,pd} + \beta_1 S_{mpd}^{act,\,within} + \beta_2 \bar{S}_{d}^{act,\,between} + \beta_3 R_{mpd} + \beta_4 \left(S_{mpd}^{\text{act,within}} \times R_{mpd}\right) + \varepsilon_{mpd},$$

with the same random-intercept structure as in Step 1a.

The effect of momentary actual similarity on love is thus:

$$\frac{\partial L_{mpd}}{\partial S_{mpd}^{act,within}} = \beta_1 + \beta_4\, R_{mpd}.$$

Accordingly, β_1 represents the momentary similarity–love association when $R_{mpd} = 0$ (i.e., at the person's typical level of perceived partner responsiveness), whereas β_4 captures how much the similarity–love slope changes for a one-unit increase in momentary perceived responsiveness relative to that person's typical level.

For perceived similarity, a parallel moderation model will be estimated, with the same centering, decomposition, and random-intercept structure:

$$L_{mpd} = b_{0,pd} + \beta_1 S_{mpd}^{perc,\,within} + \beta_2 \bar{S}_{pd}^{perc,\,between} + \beta_3 R_{mpd} + \beta_4 \left(S_{mpd}^{\text{perc,within}} \times R_{mpd}\right) + \varepsilon_{mpd},$$

For simplicity, we use the same coefficient symbols across the two models; coefficients are estimated separately for the actual- and perceived-similarity models.

For PA similarity, H3.1 states that perceived partner responsiveness amplifies positive effects, meaning that the similarity–love slope in both models becomes more positive at higher perceived partner responsiveness, also corresponding to $\beta_4 > 0$. For NA similarity, H3.2 predicts that perceived responsiveness buffers negative effects, meaning that the (typically negative) similarity–love slope becomes less negative at higher perceived partner responsiveness. This corresponds to $\beta_4 > 0$ (increasing R_{mpd} then makes $\beta_1 + \beta_4 R_{mpd}$ larger). Where interactions are significant, we will probe simple slopes at low (−1 SD), average, and high (+1 SD) levels of partner responsiveness.

Step 3b: Moderation by event valence (H3.3–H3.4)

To test the context moderation hypotheses (H3.3–H3.4), we construct a momentary context index (positive vs. negative events since the previous beep). At each beep, participants report whether something positive and something negative happened since the previous beep (1 = no, 2 = yes, a little, 3 = yes, a lot). For each person p in dyad d at moment m, we will recode these items to a 0–2 scale (0 = no, 1 = a little, 2 = a lot). We will then construct a momentary context index

$$C_{mpd} = Positive\ Event_{mpd} - Negative\ Event_{mpd}$$

so that higher values indicate relatively more positive (vs negative) contexts and lower values indicate relatively more negative contexts. Consistent with our general centering strategy, we will person-mean center C_{mpd} so that 0 reflects a person's typical event context, $C_{mpd} > 0$ in moments more positive than typical, and $C_{mpd} < 0$ in moments more negative than typical.

For actual similarity, the moderation model extends Step 1a as

$$L_{mpd} = b_{0,pd} + \beta_1 S_{mpd}^{act,\,within} + \beta_2 \bar{S}_{d}^{act,\,between} + \beta_3 C_{mpd} + \beta_4 \left(S_{mpd}^{\text{act,within}} \times C_{mpd}\right) + \varepsilon_{mpd}$$

with the same random-intercept structure as in Step 1a.

The effect of actual momentary similarity on love is thus

$$\frac{\partial L_{mpd}}{\partial S_{mpd}^{act,within}} = \beta_1 + \beta_4\, C_{mpd}.$$

For perceived similarity, a parallel moderation model will be estimated with the same centering, decomposition, and random-intercept structure:

$$L_{mpd} = b_{0,pd} + \beta_1 S_{mpd}^{perc,within} + \beta_2 \bar{S}_{pd}^{perc,between} + \beta_3 C_{mpd} + \beta_4 \left(S_{mpd}^{\text{perc,within}} \times C_{mpd}\right) + \varepsilon_{mpd}$$

For simplicity, we use the same coefficient symbols across the two models; coefficients are estimated separately for the actual- and perceived-similarity models.

PA similarity is expected to be more beneficial in negative contexts (H3.3), so we expect the slope in both models to be more positive when C_{mpd} is lower ($C_{mpd} < 0$), so $\beta_4 < 0$. NA similarity is expected to be more harmful in positive contexts (H3.4), so we expect the slope in both models to be more negative when C_{mpd} is higher ($C_{mpd} > 0$), and thus also $\beta_4 < 0$. We will probe simple slopes at relatively negative (−1 SD), average, and relatively positive (+1 SD) context levels.

<u>For Study Component 2 (on lab interaction data)</u>

We organize Component 2 analyses in the same stepwise structure as Component 1: (1a) estimate models with actual similarity, (1b) add perceived similarity, (1c) examine exploratory joint/conditional effects, (2) test mediation, and (3a) test moderation by trait we-ness and (3b) by interaction context. Steps 1a–1c, Step 2, and Step 3a are estimated separately for the negative and positive interaction contexts. Step 3b pools both contexts and includes interaction context as a moderator to formally test context differences.

General model specifications

Estimation Strategy. Dyads are again treated as indistinguishable. Residuals are assumed independent. Unlike Component 1, we do not specify AR(1) residual structures because the outcomes in Component 2 are at the interaction level. All models are estimated separately for PA and NA and separately for the two outcomes (love and closeness). Furthermore, we adopt a two-stage approach regarding the interaction context: For H1, H2, H3.1, and H3.2, we estimate models separately within each interaction context (positive and negative). This allows us to evaluate the robustness of the findings across different affective settings and to descriptively compare the patterns of results. For H3.3 and H3.4, we first compare contexts descriptively and then test context differences formally in pooled models using a context moderator.

Notation. Notation follows the approach as in Component 1. We introduce an additional index c for the interaction context ($c = 0$ negative interaction, $c = 1$ positive interaction). For each dyad d, we have two interactions c. After each interaction, each person p reports post-

interaction love and closeness, generically denoted by outcome O_{pdc}. Within each interaction c, m indexes VMR segments.

Centered affect variables. Let A_{mpdc}, P^{act}_{mpdc} and P^{perc}_{mpdc} denote, respectively, person p's self-reported affect, the partner's self-reported affect (used for actual similarity), and person p's perceived partner affect (used for perceived similarity), for VMR segments m in interaction c within dyad d. We person-mean center these VMR affect ratings across all segments m and across both interactions c for each person p in dyad d, yielding centered affect variables:

$$X_{mpdc} = A_{mpdc} - \frac{1}{N_{pd}} \sum_{m,c} A_{mpdc}\,,$$

$$Y^{act}_{mpdc} = P^{act}_{mpdc} - \frac{1}{N_{pd}} \sum_{m,c} P^{act}_{mpdc}\,, \qquad Y^{perc}_{mpdc} = P^{perc}_{mpdc} - \frac{1}{N_{pd}} \sum_{m,c} P^{perc}_{mpdc},$$

where N_{pd} is the number of available VMR segments for person p in dyad d across both interactions. Centering across interactions defines deviations relative to each person's overall (positive and negative ilab-sessions combined) baseline. We will optionally re-estimate using within-interaction centering as a sensitivity check.

Similarity indices. Momentary similarity is operationalized as joint elevation in baseline-centered affect penalized for mismatch, mirroring the definition in study component 1. For each VMR segment m in interaction c within dyad d, we define:

$$S^{act}_{mpdc} = \frac{X_{mpdc} + Y^{act}_{mpdc}}{2} - |\, X_{mpdc} - Y^{act}_{mpdc} \,|,\; S^{perc}_{mpdc} = \frac{X_{mpdc} + Y^{perc}_{mpdc}}{2} - |\, X_{mpdc} - Y^{perc}_{mpdc} \,|.$$

Higher values thus indicate more shared elevated affect (both partners relatively high vs. their baselines) combined with low mismatch, whereas shared low affect and/or strong divergence yield lower values.

Actual similarity is dyad-level in the sense that it is based on both partners' centered self-reports while perceived similarity is person-specific because it depends on person p's perceptions. S^{act}_{mpdc} is symmetric in a dyad (it will be identical for both partners).

We average the momentary similarity indices across segments m within each interaction c:

$$\bar{S}^{act}_{pdc} = \frac{1}{N_{pdc}} \sum_{m} S^{act}_{mpdc}\,, \bar{S}^{perc}_{pdc} = \frac{1}{N_{pdc}} \sum_{m} S^{perc}_{mpdc}\,,$$

where N_{pdc} is the number of available VMR segments for person p in dyad d during interaction c.

Trait we-ness. Trait we-ness is denoted W^{T}_{pd} and is constructed from the Dyadic Coping Inventory (DCI) as the mean of the DCI items that reflect positive, shared dyadic coping (see section on Measured Variables). We will report internal consistency (e.g., Cronbach's α, McDonalds ω) for this trait we-ness composite. We again grand-mean center W^{T}_{pd} so that 0 reflects the average level of trait we-ness in the sample.

Random intercepts and estimation. Since the models in Steps 1a–1c, Step 2, and Step 3a are estimated separately for the positive and negative interaction contexts, we will include only a random intercept for dyads (to account for the interdependence between partners). Because these models contain only one observation per person, we cannot add a person-level random intercept.

For the context moderation analysis (Step 3b), we pool data from both interaction contexts (yielding two observations per person). We will thus include random intercepts for both dyads and persons nested within dyads. All models are estimated separately for PA and NA and separately for the two outcomes (love and closeness).

Step 1a: Actual similarity (H1.1-H1.2)

For both interaction types c, we estimate:

$$O_{pdc} = b_{0,d} + \beta_1 \bar{S}^{\text{act}}_{pdc} + \varepsilon_{pdc},$$

with random intercept

$$b_{0,d} = \beta_0 + v_{0,d},$$

where β_0 is the fixed intercept, $v^{act}_{0,d} \sim N(0, \tau^2_d)$ is the dyad-level random intercept and $\varepsilon_{pdc} \sim N(0, \sigma^2)$ is the residual.

Here, β_1 is the association between interaction-level actual similarity and the post-interaction outcome. For PA similarity (H1.1), we expect this coefficient to be positive, while for NA similarity (H1.2), we expect this to be negative.

Step 1b: Add perceived similarity (H1.3)

Next, we add interaction-averaged perceived similarity alongside actual similarity (without an interaction term) to evaluate whether perceived similarity predicts post-interaction evaluations above and beyond actual similarity, and whether the actual similarity coefficient is reduced in magnitude when perceived similarity is included.

$$O_{pdc} = b_{0,d} + \beta_1 \bar{S}^{act}_{pdc} + \beta_2 \bar{S}^{perc}_{pdc} + \varepsilon_{pdc},$$

with the same random intercept as in Step 1a.

Evidence consistent with H1.3: β_2 in the expected direction, and attenuation of β_1 relative to Step 1a.

In this additive model β_1 quantifies the association between actual similarity and the post-interaction evaluation, holding perceived similarity constant. β_2 quantifies the unique association between perceived similarity and the evaluation, above and beyond actual similarity. A significant β_2 indicates that ‘feeling similar’ predicts relationship evaluations even after accounting for objective alignment.

We will interpret the results as supporting H1.3 if β_2 is significant and in the expected direction (positive for PA, negative for NA) and the effect of actual similarity is attenuated relative to Step 1a.

Step 1c (Exploratory): Actual × perceived interaction

As an exploratory extension, we will estimate models that include both similarity indices and their interaction term. This allows us to examine whether the association of one form of similarity with post-interaction evaluations depends on the level of the other.

To ensure the main effects (β_1 and β_2 in the model below) remain interpretable as effects at the average level of the moderator, we will grand-mean center the interaction-averaged similarity indices (within each interaction context) prior to forming the product term. We define the centered predictors as $\tilde{S}_{pdc}^{act}$ and $\tilde{S}_{pdc}^{perc}$.

$$O_{pdc} = b_{0,d} + \beta_1 \tilde{S}_{pdc}^{act} + \beta_2 \tilde{S}_{pdc}^{perc} + \beta_3 \left(\tilde{S}_{pdc}^{act} \times \tilde{S}_{pdc}^{perc}\right) + \varepsilon_{pdc},$$

with the same random intercept as in Step 1a.

β_3 tests whether the strength of the association between actual similarity and the outcome depends on the level of perceived similarity (and vice versa). If this interaction is significant, we will probe simple slopes for combinations of perceived and actual similarity at low (-1 SD), average (0), and high (+1 SD) levels. Of particular interest is whether the slope of perceived similarity remains positive and significant even when actual similarity is low ('illusory' similarity) and whether actual similarity is most predictive when perceived similarity is high (i.e., when being similar also 'feels' similar).

Step 2: Mediation by perceived similarity (H2.1-H2.2)

To test H2.1 and H2.2, we estimate mediation models in which interaction-averaged perceived similarity mediates the association between interaction-averaged actual similarity and O_{pdc}.

We estimate the mediator model:

$$\bar{S}_{pdc}^{perc} = t_{0,d} + \theta_1 \bar{S}_{pdc}^{act} + \varepsilon_{pdc}^{M},$$

with random intercept

$$t_{0,d} = \theta_0 + v_{0,d}^{M},$$

where θ_0 is the fixed intercept, $v_{0,d}^{M}$ is the dyad-level random intercept, and ε_{pdc}^{M} is the residual.

The outcome model is:

$$O_{pdc} = f_{0,d} + \varphi_1 \bar{S}_{pdc}^{perc} + \varphi_2 \bar{S}_{pdc}^{act} + \varepsilon_{pdc}^{O},$$

with random intercept

$$f_{0,d} = \varphi_0 + v_{0,d}^{O},$$

where φ_0 is the fixed intercept, $v_{0,d}^{O}$ is the dyad-level random intercept, and ε_{pdc}^{O} is the residual.

The direct effect of interaction-level actual similarity on the outcome is quantified by φ_2, while the indirect effect via perceived similarity is given by $\theta_1 \varphi_1$. For H2.1 (PA), we expect $\theta_1 > 0$

(more actual similarity predicts more perceived similarity) and $\varphi_1 > 0$ (more perceived similarity predicts more positive post-interaction evaluations) and hence a positive indirect effect $\theta_1\varphi_1 > 0$. We will explore whether there is a direct effect ($\varphi_2 > 0$), which would suggest partial mediation. For H2.2 (NA) we expect $\theta_1 > 0$ but $\varphi_1 < 0$, yielding a negative indirect effect. We will also explore whether there is a direct effect ($\varphi_2 < 0$).

We will obtain confidence intervals for the indirect effect using the same Monte Carlo simulation approach for multilevel mediation as in Component 1, but now applied to these interaction-level variables.

Step 3a: Moderation by trait we-ness W_{pd}^T

To test H3.1–H3.2, we examine whether the interaction-level similarity–evaluation association depends on a person's trait we-ness W_{pd}^T. Consistent with our general centering strategy, W_{pd}^T will be grand-mean centered so that 0 reflects the sample-average level of trait we-ness.

Because H3.1–H3.2 are formulated for both actual and perceived similarity, we estimate parallel moderation models. For simplicity, we use the same coefficient symbols across the two models. Coefficients are estimated separately for the actual- and perceived-similarity models. We estimate these models separately within the positive interaction and within the negative interaction.

For actual similarity, we estimate

$$O_{pdc} = b_{0,d} + \beta_1 \bar{S}_{pdc}^{act} + \beta_2 W_{pd}^T + \beta_3 \left(\bar{S}_{pdc}^{act} \times W_{pd}^T\right) + \varepsilon_{pdc}$$

with the same random intercept as in Step 1a.

For perceived similarity, we estimate

$$O_{pdc} = b_{0,d} + \beta_1 \bar{S}_{pdc}^{perc} + \beta_2 W_{pd}^T + \beta_3 \left(\bar{S}_{pdc}^{perc} \times W_{pd}^T\right) + \varepsilon_{pdc}$$

with the same random-intercept structure.

In both models, β_3 tests whether the similarity–evaluation association depends on trait we-ness. The conditional association between similarity and the post-interaction outcome is $\beta_1 + \beta_3 W_{pd}^T$. For PA similarity (H3.1), we expect higher trait we-ness to amplify positive effects, such that $\beta_3 > 0$ (the slope actual or perceived similarity on O_{pdc} becomes more positive at higher W_{pd}^T). For NA similarity (H3.2), we expect higher trait we-ness to buffer negative effects, such that $\beta_3 > 0$ (the typically negative slope becomes less negative at higher W_{pd}^T).

For interpretability, we will further probe significant interactions by estimating simple slopes of similarity on O_{pdc} at low(-1 SD), average (0), and high ($+1$ SD) levels of W_{pd}^T, analogous to the simple-slope procedures described for Component 1.

Step 3b: Context moderation (negative vs positive interaction)

In Component 2, situational context is manipulated directly via interaction valence: couples complete a positive interaction task and a negative interaction task. This allows a test of H3.3-H3.4 in which context is defined by the interaction.

We examine moderation by interaction valence in two complementary ways for both actual and perceived similarity:

- **Descriptive comparison (stratified models):** First, we will estimate the Step 1a model separately for the negative and positive interaction and descriptively compare the similarity–evaluation coefficients for actual similarity. To examine the same comparison for perceived similarity, we will estimate analogous models as Step 1a, substituting perceived similarity for actual similarity, separately within each interaction type.
- **Formal test:** Second, we define a dummy variable $\boldsymbol{C_{pdc} = c}$ ($\boldsymbol{c = 0}$ negative interaction, $\boldsymbol{c = 1}$ positive interaction). We will then pool all interaction observations and include $\boldsymbol{C_{pdc}}$ and its interaction with similarity as predictors. This provides a direct statistical test of whether the similarity–evaluation association differs between positive and negative interactions.

For actual similarity, we estimate:

$$O_{pdc} = b_{0,pd} + \beta_1 \bar{S}^{\text{act}}_{pdc} + \beta_2 C_{pdc} + \beta_3 (\bar{S}^{\text{act}}_{pdc} \times C_{pdc}) + \varepsilon_{pdc},$$

with random intercept structure

$$b_{0,pd} = \beta_0 + v_{0,d} + u_{0,pd},$$

where β_0 is the fixed intercept, $v_{0,d}$ is the dyad-level random intercept, $u_{0,pd}$ is the person-level random intercept, and ε_{pdc} is the residual. Note that because this model pools two observations per person, we re-introduce the person-level random intercept.

For perceived similarity, we estimate:

$$O_{pdc} = b_{0,pd} + \beta_1 \bar{S}^{perc}_{pdc} + \beta_2 C_{pdc} + \beta_3 (\bar{S}^{perc}_{pdc} \times C_{pdc}) + \varepsilon_{pdc},$$

with the same random intercept structure. For simplicity, we use the same coefficient symbols across the two models with coefficients estimated separately for the actual- and perceived-similarity models.

In this parameterization, β_1 is the similarity–evaluation slope for the negative interaction ($C_{pdc} = 0$), and $\beta_1 + \beta_3$ is the slope for the positive interaction **(**$C_{pdc} = 1$).

For PA similarity (H3.3), we expect the positive similarity–evaluation association to be stronger in the negative interaction than in the positive interaction. This implies $\beta_3 < 0$ (the slope in the positive interaction is less positive than in the negative interaction). For NA similarity (H3.4), we expect the negative association with O_{pdc} to be stronger (more negative)

in the positive interaction than in the negative interaction. Again, this corresponds to $\beta_3 < 0$ (the slope becomes more negative when $C_{pdc} = 1$).

For interpretability, we will compute and report simple slopes of similarity separately for positive and negative interactions (i.e., at $C_{pdc} = 0$ and $C_{pdc} = 1$) and their confidence intervals.

Transformations

We will person-mean center Level-1 predictors in all models and grand-mean center Level-2 predictors.

Inference criteria

We will use p <.05 as the criterion for a significant effect using a two tailed test in all models. We will also report confidence intervals. We will not correct for multiple comparisons since each model reflects a different hypothesis (García-Pérez, 2023).

Data exclusion

For study component 1, we will exclude one couple because, due to technical issues, it is uncertain which ESM questionnaire responses belong to which partner. This couple also did not complete the lab session. We will also exclude couples where at least one of the partners has an ESM compliance ratio (number of completed beeps over number of received beeps) of less than 0.30.

Missing data

The general principle behind the treatment of missing data is:

- Actual similarity at a moment/segment can only be computed when both partners provided the relevant affect reports for that same moment/segment.
- Perceived similarity is actor-reported, so it can be computed even if the partner did not respond, but any analysis that includes actual similarity (or compares/mediates between actual and perceived) necessarily relies on the subset of matched moments/segments where both partners responded.

ESM

We will not impute missing intensive longitudinal observations. For all primary analyses, we will analyze only those moments for which the actual similarity index can be computed, i.e., moments where both partners provided the required affect reports at the same moment and the actor provided the outcome. Moments where either partner is missing the required affect

report will be excluded because the dyadic actual similarity index cannot be computed. For analyses that additionally include perceived similarity (e.g., unique effects and mediation models), we will further restrict to moments where the perceived similarity index is available. There will typically be no such further restriction, except in cases of item-level missingness.

Lab interactions / VMR

We do not expect substantial missing data because VMR questions were compulsory to answer. If missingness would occurs due to technical issues, we will exclude the affected segments from analyses, while retaining the remaining segments for that person/dyad.

Background questionnaire (DCI)

Trait we-ness will be computed as the mean of selected DCI items reflecting positive/shared dyadic coping. Items that were not implemented in Qualtrics (structural missingness; i.e., missing for all participants) will not be used.

Exploratory analysis

Study Component 1

As an exploratory extension to the component 1 H1.3 model, we will additionally include an interaction between actual and perceived similarity to test whether the association of each form of similarity on momentary love is conditional on the level of the other.

- **Exploratory (conditional effects and probing).** In these models, we will probe predicted love for combinations of low (–1 SD), average, and high (+1 SD) values of actual and perceived similarity. This allows us to evaluate (a) whether perceived similarity is associated with higher love when actual similarity is low (consistent with some kind of 'illusory' benefit), and (b) whether actual similarity is most predictive when perceived similarity is high (i.e., when being similar also 'feels' similar).

In the component 1 H2 mediation hypothesis, we expect perceived similarity to play a central mediating role but will treat any residual direct path from actual similarity to love as an open, more exploratory question.

- **Exploratory (direct effect).** We will also explore whether actual similarity retains a direct association with love after accounting for perceived similarity (indicating partial rather than full statistical mediation), consistent with the idea that both 'really being similar' and 'feeling similar' matter.

Study Component 2

As an exploratory extension to the component 2 H1.3 model, we will additionally include an interaction between interaction-averaged actual similarity and interaction-averaged perceived similarity to test whether the association of each form of similarity with post-interaction evaluations is conditional on the level of the other.

OSF Preregistration

- **Exploratory (conditional effects and probing).** If the interaction term is significant, we will probe predicted post-interaction love and closeness for combinations of low (–1 SD), average, and high (+1 SD) levels of actual and perceived similarity. This allows us to evaluate (a) whether perceived similarity is associated with more positive evaluations even when actual similarity is low (consistent with an 'illusory' benefit of feeling aligned), and (b) whether actual similarity is most predictive when perceived similarity is high (i.e., when being similar also feels similar).

In the component 2 H2 mediation hypothesis, we expect perceived similarity to play a central mediating role but will treat any residual direct path from actual similarity to love or closeness as an open, more exploratory question.

- **Exploratory.** We will also explore whether average actual similarity retains a direct association with post-interaction evaluations after accounting for average perceived similarity (partial vs full statistical mediation).

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